Curriculum Vitae

Personal Information

Gauri Vaidya, PhD
University of Limerick
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Last Updated: October 21, 2026

Education

2021–2025 **Ph.D., Artificial Intelligence**
University of Limerick, Ireland
Thesis Title: Efficient Hyperparameter Optimization for Streamlining Model Training in Machine and Deep Learning
Supervisors: Prof. Conor Ryan and Prof. Meghana Kshirsagar

2016–2020 **Bachelors of Engineering, Computer Science and Engineering**
Government College of Engineering, Aurangabad, India
Ranking: 2nd of 70 students (A++)
Thesis Title: Blockchain-based systems for secure handling of first information reports
Supervisor: Dr. Vivek Kshirsagar

Professional Experience

2025-Present **National Kidney Disease Surveillance System (NKDSS)**
University of Limerick, Ireland
Position: Level 2 Postdoctoral Researcher
Supervisors: Prof. Austin Stack
Contributions: (1) Applying advanced data analysis and machine learning methods to quantify the burden and impact of chronic kidney disease (CKD) in Ireland, supporting evidence-based prevention and intervention strategies for improved patient outcomes. (2) Developing a secure and linked health data infrastructure integrating multiple national sources (e.g., hospital, registry, and population data) to track CKD prevalence, risk factors, and quality of care across Ireland.

2021-2025 **Biocomputing and Developmental Systems Research Group**
University of Limerick, Ireland
Position: PhD Researcher
Supervisors: Prof. Conor Ryan and Prof. Meghana Kshirsagar
Contributions: (1) Developed three novel hyperparameter optimization frameworks (HyperEstimator, HyperGE, PurGE), achieving up to 2.2× faster convergence, 40% energy savings, and 30% lower memory usage compared to state-of-the-art methods.
(2) Co-authored 10 peer-reviewed research publications during the PhD and through multiple collaborations; presented work nationally and internationally via invited talks, technical presentations, and poster sessions.
(3) Served as a Teaching Assistant for Computer Science modules and mentored undergraduate, master's students, and research interns on applied machine learning, dissertations, and project development. (4) Managed the NVIDIA A100 GPU server for the department of Computer Science and Information Systems, overseeing access control, resource allocation, and providing technical support for high-performance AI workloads.

2019–2021 **Biocomputing and Developmental Systems Research Group**
Position: Research Intern
Supervisors: Prof. Conor Ryan and Dr. Meghana Kshirsagar
Contributions: (1) Developed a cryptographically secure pseudo-random number generator in collaboration with Intel, validated against NIST SP800-22 standards, resulting in a peer-reviewed publication in Nature Scientific Reports.
(2) Built a blockchain-based framework for secure electronic healthcare data sharing using Ethereum and Neo4j, deployed under the EU Horizon 2020 project, with outcomes published in Frontiers in Blockchain.

2020–2021 **Principal Global Services**
Position: Trainee Analyst
Manager: Shridhar More
Contributions: (1) Automated key workflows in insurance operations using full-stack development within the Salesforce framework, leading to a 20% improvement in process efficiency.
(2) Recognized for strong commitment and technical contributions, and invited to join the core development team within 6 months, where I contributed to feature development and collaborated closely with cross-functional teams internationally.

RESEARCH PROJECTS

2025-Present **Model Evolution and Explainability Analysis Using Real-World Evidence in Lung and Breast Cancer**

- Ongoing work as part of an international working group, utilizing a large, previously unpublished real-world EMR dataset of over 10,000 lung and breast cancer patients with detailed clinical, treatment, demographic, and limited omics data.
- Designing an iterative ML framework to investigate how model reproducibility, complexity,

and explainability evolve with increasing training data size, and to assess the traceability of shifting patient-level classifications over time.

- Aiming to inform robust, transparent clinical decision-making by understanding how real-world model performance adapts to longitudinal data growth and diverse global patient cohorts.

2025-Present **Artificial Intelligence Powered Decision Support System for Sustainable Food Packaging of Breakfast Food Items**

- Developed an AI-powered diagnostic system using ResNet-50 and DeepHyperX spectral features to classify four common plastic types in breakfast packaging, assessing potential cancer risks.
- Integrated demographic and regional cancer data from WHO/GLOBOCAN to generate personalized risk scores, enhancing model explainability and public health relevance.
- Deployed an interactive Streamlit interface for real-world user engagement, enabling informed consumer choices and supporting sustainability and food safety policy efforts.

2024-Present **Mitigating Algorithmic Bias in Prostate Cancer Risk Stratification with Responsible AI and ML**

- Designed a dual-modality ML pipeline using logistic regression on clinical features and CNN-based image triage (ResNet50, Inception v3) to stratify clinically significant prostate cancer (csPCa) without relying on invasive biopsy-derived features.
- Implemented bias mitigation strategies, including demographic-aware dataset slicing, SMOTE for class balancing, and MRI-sequence mixing to reduce confirmation, protocol, and equipment bias in model training.
- Achieved 69.79% test accuracy on clinical data and 60% on MRI-based classification, enhancing model trustworthiness and demonstrating the viability of responsible AI in equitable PCa risk stratification.

2023-Present **Reducing overcrowding in Irish Emergency Departments**

- Conducting ongoing research to optimize patient flow for hip fracture cases by integrating AI and agent-based modelling (ABM) within the University Hospital Limerick group using real-world administrative and clinical datasets.
- Applying machine learning and simulation tools to identify system bottlenecks, predict service demand, and evaluate targeted interventions, with a focus on improving compliance with key healthcare performance indicators (e.g., time-to-ward, time-to-theatre).
- Aiming to reduce overcrowding and improve healthcare delivery by creating an interpretable, data-driven dashboard and forecasting framework for operational decision-making in Irish acute care environments.

2024-Present **AI-Driven Patient Stratification and Outcome Modeling for Chronic Lymphocytic Leukemia in Ireland**

- Engineered a multimodal knowledge graph using Neo4j and schema-based modeling to integrate genomic datasets (cBioPortal), clinical trial registries (ClinicalTrials.gov), and national treatment guidelines (NCCP) to support advanced CLL care in Ireland.

- Developed a precision medicine pipeline by combining retrieval-augmented generation (RAG) with fine-tuned LLMs (e.g., Flan-T5, GPT-4), grounding outputs in structured graph data to enhance interpretability and reduce hallucinations in clinical recommendations.
- Applied overlapping community detection algorithms to uncover non-obvious patient subgroups within Irish cohorts, revealing sociocultural and systemic patterns influencing treatment decisions and healthcare access.

2021-2025 **Efficient Hyperparameter Optimization for Streamlining Model Training in Machine and Deep Learning**

- Designed three automated frameworks to streamline hyperparameter tuning in machine learning pipelines, achieving up to 2.2× faster convergence, 40% energy savings, and 30% lower memory usage.
- Executed 50+ experiments and ablation studies with experiment tracking on EfficientNetB7, ResNet50, VGG19, and XGBoost to evaluate trade-offs between accuracy, training time, and resource consumption.
- Benchmarked frameworks against Random Search, Grid Search, TPE, and CMA-ES, consistently outperforming baselines in tuning efficiency and accuracy across deep learning and tabular ML tasks.

2022-2023 **Predictive Analytics for Transport Decarbonization in Ireland**

- Built a 3-phase machine learning framework combining regression and time-series forecasting to assess CO_2 emissions, predict EV adoption, and estimate renewable energy capacity for powering EVs in Ireland.
- Achieved an R^2 of 0.86 and RMSE of 0.36 using a Random Forest model to estimate CO_2 emissions based on car type, year, and engine size—enabling dynamic carbon tax estimation based on vehicle age and emissions profile.
- Forecasted a 54.41% increase in EV adoption using SARIMA and ANN models on national registration data; captured seasonal spikes caused by Ireland's dual vehicle registration policy.
- Demonstrated that 300 wind farms could power 26,700 electric vehicles daily, using ANN models trained on 10-minute wind energy data with R2 of 0.918 and RMSE of 25.48, demonstrating the feasibility of clean transport energy procurement.

2019-2022 **Scalable Blockchain Framework for Healthcare Data Sharing**

- Developed a secure, patient-controlled data sharing system using Ethereum, Multichain, and Web3j, enabling private, tamper-proof exchange of medical records across hospitals, labs, insurers, and pharmacies.
- Integrated agent-based smart contracts with Neo4j and RDF-based knowledge graphs, enabling real-time, role-specific access and semantic search across multimodal health data.
- Built predictive analytics dashboards and data pipelines using Python, Flask, and Synthea, allowing healthcare providers to monitor key metrics, forecast outbreaks, and optimize care delivery workflows.

2022-2024 **Single-cell and knowledge graphs in differential expression profiling from single-cell RNA transcriptomic data**

- Processed and annotated single-cell RNA seq datasets from gastric and head & neck squamous cell carcinoma using Seurat, with SingleR and scMRMA for cell-to-cluster assignment, to extract immune cell populations and cohort-specific gene expression profiles.
- Built cohort-level knowledge graphs from annotated scRNA-seq data to represent biomarker–cell type–patient relationships, enabling exploration of shared and distinct signatures across cancer stages and immune subtypes.
- Enabled Cypher-based querying to analyze differentially expressed immune-related genes and investigate their role in cellular heterogeneity, and cohort analysis.

2019-2022 **Cryptographically Secure Random Number Generator with Bio-inspired Machine Learning**

- Designed a cryptographically secure pseudo-random number generator using bio-inspired machine learning approach to generate high-entropy seeds; achieved an average 8-bit entropy of 7.94, with potential to generate 1.5M+ unique seeds.
- Passed all 15 NIST SP800-22 and 17 Diehard tests, demonstrating statistical robustness and resistance to attacks such as seed prediction, repetition, and reverse engineering; leveraged SHA3-512 for secure hashing of evolving seeds.
- Benchmarked the tool against Python's rand() via Monte Carlo simulations and encryption throughput analysis, outperforming baselines in estimating π (error < 0.005) and achieving faster encryption speeds across RGB image datasets.

FUNDING

2023–2025 **Unleashing the power of AI and Digital Twins in Emergency Care:A Vision for the Future (ALTER)**
Funding Body: Research Ireland, National Challenge Fund call, Digital for Resilience
Value: €220,895
PI: Meghana Kshirsagar, co-PI: Alison O'Connor.
Role: Named Collaborator

2023–2025 **Artificial Intelligence and Digital Twins in Emergency Care: A Vision for the Future**
Funding Body: EuroHPC National Competence Centre (EuroCC) run by Irish Centre for High-End Computing (ICHEC)
Value: €70,000
PI: Alison O'Connor, co-PI: Meghana Kshirsagar.
Role: Named Collaborator

AWARDS

Apr 2023 Runner-Up, Thesis in Three (Faculty Heat), Science and Engineering Department, University of Limerick, Ireland. (3rd place among 12 participants)

Mar 2019	First Prize (€570 cash award), Smart India Hackathon for Best IoT-Based Waste Management System, India along with Rushikesh Jachak, Purva Chaudhari, Mayur Rahangdale and Prathamesh Chaudhari. (1st place among 20 finalists)
Mar 2013	Best Outgoing Student, St. Xavier's High School, India. (Selected among 200 students)

SCHOLARSHIPS

2021-2025	Doctoral Fellowship by the Research Ireland Centre for Training in Artificial Intelligence. € 22,000/year plus conference travel, training and equipments
2016-2020	Undergraduate Fellowship by Vidyadaan Sahayyak Mandal, Thane, India. Rs.50000/year.

TEACHING EXPERIENCE

Spring 2026	**Instructor** for CS6472, graduate course on "Research Methods and Specifications", University of Limerick, Ireland; sole responsibility for the course with 70 registered students.
Spring 2025	**Instructor** for CS6472, graduate course on "Research Methods and Specifications", University of Limerick, Ireland; sole responsibility for the course with 36 registered students.
Spring 2025	Teaching assistant for CS4136, undergraduate course on "Topics in AI", University of Limerick; taught by Prof. Meghana Kshirsagar.
Fall 2024	Teaching assistant for CS4207, undergraduate course on "Advanced Programming Concepts and Practices", University of Limerick; taught by Prof. Meghana Kshirsagar.
Spring 2024	Teaching assistant for CS4222, undergraduate course on "Software Development", University of Limerick; taught by Prof. Meghana Kshirsagar.
Spring 2024	Teaching assistant for CS4116, undergraduate course on "Software Development Project", University of Limerick; taught by Prof. Conor Ryan.
Spring 2024	**Moderator** for CS6163, graduate course on "Advanced Topics Seminars and Project", University of Limerick; taught by Dr. Tabea DeWille.
Fall 2023	Teaching assistant for CS4221, undergraduate course on "Foundations of Computer Science", University of Limerick; taught by Prof. Conor Ryan and Prof. Meghana Kshirsagar.
Fall 2023	Teaching assistant for CS4221, undergraduate course on "Foundations of Computer Science", University of Limerick; taught by Prof. Conor Ryan.
Spring 2023	Teaching assistant for CS4116, undergraduate course on "Software Development Project", University of Limerick; taught by Prof. Conor Ryan.
Spring 2022	Teaching assistant for CS4116, undergraduate course on "Software Development Project", University of Limerick; taught by Prof. Conor Ryan.
Fall 2022	Teaching assistant for CS4221, undergraduate course on "Foundations of Computer Science", University of Limerick; taught by Prof. Conor Ryan.

Fall 2022 **Module Leader (Online)** for undergraduate course on "Blockchain Technologies ", Maharashtra Institute of Technology, India; sole responsibility for the course with 65 registered students.

SUPERVISION

Undergraduate students

2024-2025 Primary supervisor for full-time undergraduate student Rahul Alam at University of Limerick. Thesis Title: *Ethnicity–Aware Diabetes Risk Assessment Using Machine Learning.*

Masters students

2026 Primary supervisor, Joseph Burke, Masters in Machine Learning and Artificial Intelligence, University of Limerick.

2025 Secondary supervisor, Ivor DSouza, Masters in Software Development, University of Limerick. Primary Supervisor: *Prof. Meghana Kshirsagar*

2025 Secondary supervisor, Mahika Jaguste, Masters in Software Development, University of Limerick. Primary Supervisor: *Prof. Meghana Kshirsagar*

Research Interns

2022-2023 Technical Mentor, Shravani Rajguru, Hrushabh Kale, Prithviraj Chavan, Nishant Shanmugam, Government College of Engineering, Aurangabad, India. Project Title: *DECART: Planning for Decarbonizing Transport Sector with Predictive Analytics-An Irish Case Study*. Primary Supervisor: *Prof. Meghana Kshirsagar*

2021-2022 Technical Mentor, Vedant Karmalkar, Abhishek Hole, Kunika Kulkarni, Mayuresh Khemnar, Maharashtra Institute of Technology, Aurangabad, India. Project Title: *Blockchain-powered, knowledge-driven data-sharing framework for electronic health records.* Primary Supervisor: *Prof. Meghana Kshirsagar*

PUBLICATIONS

In progress

1. **Book:** Shiny Apps for Patient Flow and Outcome Analysis with MIMIC-III
2. **Journal Article:** Artificial Intelligence Powered Decision Support System for Sustainable Food Packaging of Breakfast Food Items
3. **Journal Article:** Prime-Generating Symbolic Regression through Grammar-Guided Search Space Pruning

Journal Publications

1. KSHIRSAGAR, M., VAIDYA, G., YAO, Y., KASAR, S., AND RYAN, C. Sensible: sequestered and synergistic blockchain ecosystem. *Engineering Reports 5*, 7 (2023), e12586
2. VAIDYA, G., KSHIRSAGAR, M., AND RYAN, C. Grammatical evolution-driven algorithm for efficient and automatic hyperparameter optimisation of neural networks. *Algorithms 16*, 7 (2023), 319
3. YAO, Y., KSHIRSAGAR, M., VAIDYA, G., DUCRÉE, J., AND RYAN, C. Convergence of blockchain, autonomous agents, and knowledge graph to share electronic health records. *Frontiers in Blockchain 4* (2021), 661238
4. RYAN, C., KSHIRSAGAR, M., VAIDYA, G., CUNNINGHAM, A., AND SIVARAMAN, R. Design of a cryptographically secure pseudo random number generator with grammatical evolution. *Scientific reports 12*, 1 (2022), 1–10
5. KSHIRSAGAR, M., GUPT, K. K., VAIDYA, G., RYAN, C., SULLIVAN, J. P., AND KSHIRSAGAR, V. Insights into incorporating trustworthiness and ethics in ai systems with explainable ai. *International Journal of Natural Computing Research (IJNCR) 11*, 1 (2022), 1–23

Conferences

4. VAIDYA, G., KSHIRSAGAR, M., AND RYAN, C. Purge: Towards responsible artificial intelligence through sustainable hyperparameter optimization. *Proceedings of 17th International Conference on Agents and Artificial Intelligence* (2025)
5. KSHIRSAGAR, M., SONTAKKE, M., VAIDYA, G., ALKHAN, A., KILLEEN, A., AND RYAN, C. Mitigating algorithmic bias in prostate cancer risk stratification with responsible artificial intelligence and machine learning. *Proceedings of 17th International Conference on Agents and Artificial Intelligence* (2025)
6. O'CONNOR, A. N., RYAN, S. E., VAIDYA, G., HARFORD, P., AND KSHIRSAGAR, M. Hip fracture patient pathways and agent-based modelling. *arXiv preprint arXiv:2410.12804* (2024)
7. KSHIRSAGAR, M., VAIDYA, G., RAJGURU, S., JADHAV, P., KALE, H., SHANMUGAM, N., RYAN, C., AND KSHIRSAGAR, V. Decart: Planning for decarbonising transport sector with predictive analytics-an irish case study. In *SMARTGREENS* (2022), pp. 157–164
8. VAIDYA, G., ILG, L., KSHIRSAGAR, M., NAREDO, E., AND RYAN, C. Hyperestimator: Evolving computationally efficient cnn models with grammatical evolution. In *ICSBT* (2022), pp. 57–68
9. BINDRA, P., KSHIRSAGAR, M., RYAN, C., VAIDYA, G., GUPT, K. K., AND KSHIRSAGAR, V. Insights into the advancements of artificial intelligence and machine learning, the present state of art, and future prospects: Seven decades of digital revolution. In *Smart Computing Techniques and Applications: Proceedings of the Fourth International Conference on Smart Computing and Informatics, Volume 1* (2021), Springer Singapore, pp. 609–621
10. NAGORI, M., PATIL, A., DESHMUKH, S., VAIDYA, G., RAHANGDALE, M., KULKARNI, C., AND KSHIRSAGAR, V. Mutichain enabled ehr management system and predictive analytics. In *Smart Trends in Computing and Communications: Proceedings of SmartCom 2019* (2020), Springer Singapore, pp. 179–187

Book Chapters

4. YAO, Y., KSHIRSAGAR, M., VAIDYA, G., LIU, J., ZHANG, Y., AND RYAN, C. Blockchain-based knowledge graph for high-impact scientific collaboration networks. 2024
5. VAIDYA, G., BINDRA, P., KSHIRSAGAR, M., AND TAMANE, S. C. Privacy and security technologies for smart city development. In *Security and privacy applications for smart city development*. Springer International Publishing Cham, 2020, pp. 3–23

Extended Abstract / Short Papers

4. YAO, Y., KSHIRSAGAR, M., VAIDYA, G., AND RYAN, C. Using a bio-inspired model to facilitate the ecosystem of data sharing in smart healthcare. *Evo* 2021–Late-Breaking Abstracts* (2021), 25–30

Abstracts

4. KSHIRSAGAR, M., VAIDYA, G., BHAVSAR, K., AND RYAN, C. Assigning treatment regimens to irish patients in head and neck squamous cell carcinoma with large language models. In *BioC 2024* (2024)
5. KSHIRSAGAR, M., AND VAIDYA, G. Comparative analysis of annotation tools in grouping cell types from single-cell rna sequencing data. In *BioC 2023* (2023)
6. KSHIRSAGAR, M., VAIDYA, G., YE, Y., LYNCH, M., AND RYAN, C. scmarkerbase: Unleashing the power of single-cell and knowledge graphs in differential expression profiling from single-cell rna transcriptomic data. In *22nd International Conference on Bioinformatics (InCoB)* (2023)

Posters

4. GAURI VAIDYA, MEGHANA KSHIRSAGAR, C. R. Hyperge: Hyperparameter optimization of neural networks with grammatical evolution. Poster presented at University of Amsterdam Conference, 2024
5. GAURI VAIDYA, MEGHANA KSHIRSAGAR, C. R. Using grammatical evolution to estimate hyperparameters for neural networks. Poster presented at Lero Review, 2023
6. MEGHANA KSHIRSAGAR, GAURI VAIDYA, C. R. Disruptive digital transformation with sensible, sustainable digital health innovation. Poster presented at Digital Health Innovation Conference, 2022

INVITED TALKS AND PRESENTATIONS

2021-2025	**Hands on workshop on Python Development**, Thomond Community College, Ireland.
Oct 2024	**Knowledge graph driven framework for comparative analysis of cohorts from transcriptomics data**, Limerick Digital Cancer Research Centre, Ireland.
Aug 2024	**Demystifying Machine Learning**, 3-day Hands-on Machine Learning Workshop, Government College of Engineering, Aurangabad, India along with Chinmay Kulkarni.
Apr 2023	**Immune cell profiling from single cell RNA with R**, ISCB along with Prof. Meghana Kshirsagar.

Nov 2022 **SHIELD: Do it the safe way**, Pycon Ireland.

Aug 2022 **Researching, Visualizations and Creating ML Models for Climate Change**, CVT, Internship Outreach Program from Harvard along with Dr. Meghana Kshirsagar.

Aug 2022 **Artificial Intelligence in Smart Transportation**, Mahathma Gandhi Mission University, Aurangabad along with Dr. Meghana Kshirsagar.

COMMUNITY SERVICES

Organizing Commitee

1. Organizer, Breaking Barriers: Addressing Algorithmic Bias for Equitable Cancer Care Access, 2025.
2. Web Master, IEEE International Conference on Blockchain and Cryptocurrency, 2023.
3. Co-Organizer, Evathon, International Evolutionary Hackathon, 2021.
4. Organizer, Smart Aurangabad Hackathon, 2019.
5. Organizer, Workshop on Blockchain Technologies, 2018.
6. Organizer, Startup Weekend, 2018.

Reviewing

1. AAMAS 2025
2. Bioconductor (BioC 2023).
3. The European Conference on Computer Vision (ECCV 2022)

Administration

1. Website Administrator, Biocomputing and Development Systems Research Group, University of Limerick, Ireland.
2. HPC Administrator, NVIDIA A-100 GPU server, Department of Computer Science and Information Systems, University of Limerick, Ireland.

MEDIA AND OUTREACH

1. 30 Under 30 Top People in tech 2025, *Business Post*
2. Contribution to open source with *atoti*
3. Insights on Modern Computation, Blog on *Medium*

VOLUNTEERING

1. Technical Volunteer, Leveraging LLMs for Data Governance, Fidelity Investments, 21st October, 2024 (Delivered by Prof. Meghana Kshirsagar and Dr Vimal Kumar).
2. Technical Mentor, Vidyadaan Sahayyak Mandal, India. 2020-present.
3. Volunteer, Patient Public Involvement Workshop, University of Limerick, 2023.
4. Teacher and Volunteer, Share a Book India Association, India, 2019-2020.

TRAINING AND CERTIFICATION

2024	Microcredential, Introduction to Real world Data in Cancer Clinical Research
2023	Oxford Machine Learning Summer School
2019	Improving Deep Neural Networks:Hyperparameter tuning, Regularization & Optimization, Coursera
2019	Convolutional Neural Networks, Coursera

LANGUAGES

English (Professional), Hindi (Professional), Marathi (Professional)